



Menopause Hormone Therapy in Women With Cardiovascular Risk Factors or Established Ischemic Heart Disease: A Joint FIGO and WHF Position Paper

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ABSTRACT

Extending women's health-span is a key challenge for the medical community. Although women live longer than men, they face a higher burden of chronic diseases, such as cardiovascular diseases (CVDs) and disabilities.

Endogenous estrogens appear to benefit cardiovascular health, and their marked decline at menopause is associated with increased cardiovascular risk; conversely, poor cardiovascular health may hasten menopause onset.

Systemic menopause hormone therapy (MHT) begun early—ideally soon after symptom onset and at most within 10 years of the final menstrual period—alleviates climacteric symptoms and is associated with long-term benefits.

This joint position paper between the International Federation of Gynecology and Obstetrics (FIGO) and the World Heart Federation (WHF) aims at supporting both gynecologists and cardiologists in providing a tailored and evidence-based counseling on systemic MHT use to women with increased CVD risk or established ischemic heart disease (IHD).

Women with CVD risk factors require thorough risk stratification to tailor MHT type, dose, and route. For those at high thromboembolic risk, transdermal estrogen, with micronized progesterone or dydrogesterone when needed, is considered the safest regimen.

Women with established IHD may receive systemic MHT for symptom relief, provided risks and benefits are carefully reviewed. Transdermal low-dose estrogen, alone or with micronized progesterone or dydrogesterone, is the best choice. All primary and secondary prevention measures, including statins, should be continued.

Managing the most complex cases warrants the adoption of a multidisciplinary approach between cardiologists, gynecologists, internists and primary care providers to offer the best possible care.

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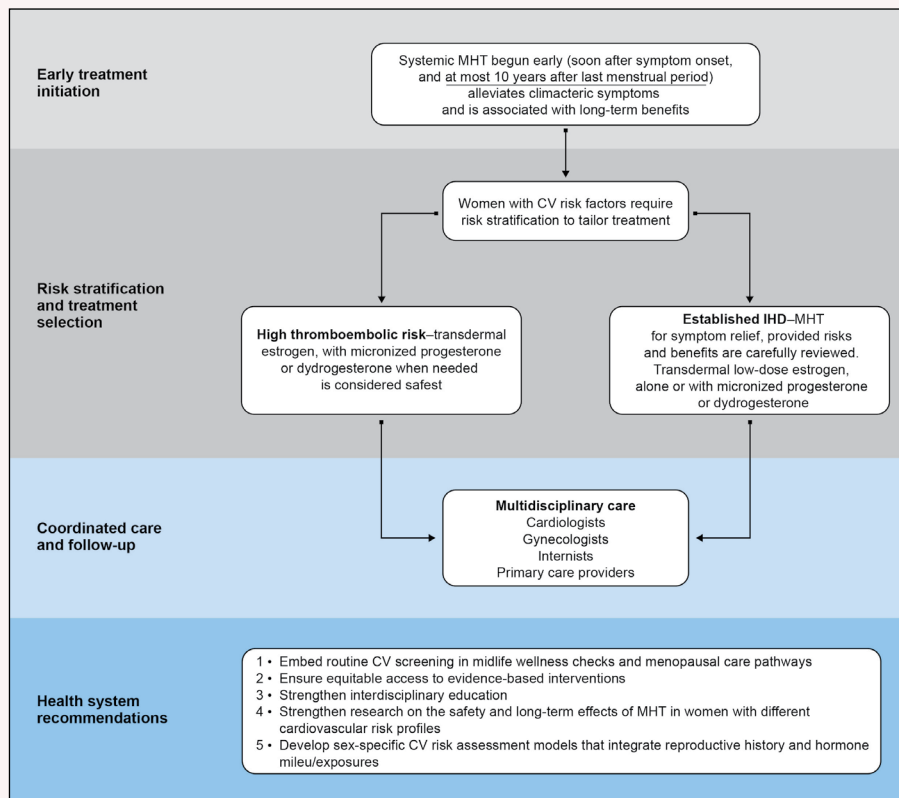
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BACKGROUND

Globally, women generally have a longer life expectancy than men; however, they also tend to spend more years living with chronic conditions and disabilities, including cardiovascular diseases (CVD), osteoporosis-related fractures, inflammatory diseases, cognitive decline and/or dementia. This poses a heavy burden both individually and on society as a whole (1). Increasing health-span is therefore one of the main challenges the medical community has to face. An appropriate management of menopause can contribute to this objective, as it positively impacts women's health in both the short- and long-terms.

Over the past 40–50 years, the use of and attitudes toward menopausal hormone therapy (MHT) among both clinicians and the general population have been profoundly shaped by the results of several pivotal scientific studies (2–6). These investigations have clarified a fundamental principle: the potential benefits of systemic MHT are highly dependent on the timing of initiation—the so-called 'timing hypothesis' (7–9). Specifically, systemic MHT appears to confer its most favorable effects when initiated early, ideally no later than 10 years after the final menstrual period and before substantial age-related changes develop across multiple organs and systems. In contrast, initiating MHT beyond this timeframe may lead to neutral or even adverse effects rather than the expected benefits. Moreover, it is now well established that also the type and the route of administration of MHT can significantly influence its overall benefit–risk profile (10).

A closer look at the historical trajectory of MHT highlights how clinical practice evolved through distinct phases shaped by emerging evidence and shifting interpretations. During the 1980s and 1990s, MHT experienced rapid expansion, driven by expectations of symptomatic relief and presumed major health benefits, including cardioprotection (2–6). This trajectory changed abruptly in 2002 with the publication of the first Women's Health Initiative (WHI) hormone therapy trial, whose initial findings reported increased risks for cardiovascular events (11).

However, many researchers have since highlighted important methodological limitations of that study, especially the advanced age of participants, with many women at enrollment more than a decade beyond menopause; the use of only one hormone regimen, while the results were initially generalized to all forms of MHT; the use of a single fixed-dose, non-individualized therapy; and high discontinuation rates, which limited the applicability of its conclusions to

younger women initiating therapy near menopause (12–19). As evidence was reinterpreted and new data emerged, clinical practice shifted toward a more nuanced, individualized approach that considers a woman's age, time since menopause, underlying risk factors, physical characteristics and current health status to ensure that MHT benefits outweigh the risks (10).

This joint FIGO and WHF paper aims at establishing a 'common language' between Gynecologists and Cardiologists to help them provide personalized counselling of women with CVD risks or established ischemic heart disease (IHD) on the use of MHT. The paper was collaboratively developed by experts, selected on the basis of their expertise in gynecology and cardiology, each contributing to different sections aligned with their scientific background. The paper contains extensive references to recent literature, including systematic reviews, meta-analyses, clinical trials and international guidelines and represents a comprehensive synthesis of available evidence, aimed at guiding clinical practice across specialties.

CARDIOVASCULAR DISEASE IN WOMEN

DEFINITIONS

CVD is a broad term that refers to a range of disorders affecting the heart and blood vessels (20). It is the leading cause of morbidity and mortality in women globally, with IHD accounting for the largest share of the CVD burden. For this reason, the present paper focuses exclusively on IHD and does not consider cerebrovascular disease or other types of heart disease. In literature, the terms coronary heart disease (CHD), IHD and coronary artery disease (CAD) are often used interchangeably (21).

The IHD spectrum includes: obstructive epicardial CAD, defined as $\geq 50\%$ stenosis of a major epicardial coronary artery on coronary angiography; myocardial infarction with nonobstructive CAD (MINOCA), defined as $< 50\%$ stenosis of a major epicardial coronary artery on coronary angiography; ischemia with nonobstructive CAD (INOCA); spontaneous coronary artery dissection (SCAD); coronary microvascular dysfunction; vasospastic angina and coronary thrombosis/embolism (Figure 1).

Furthermore, IHD presentation includes acute ST-elevation myocardial infarction (STEMI), non-ST elevation myocardial infarction (NSTEMI), unstable angina (acute coronary syndrome: ACS), chronic stable angina and ischemic cardiomyopathy (heart failure due to IHD). The term atherosclerotic cardiovascular disease (ASCVD) refers to clinical and subclinical plaque of any artery including coronary, cerebral, peripheral, and the aorta.

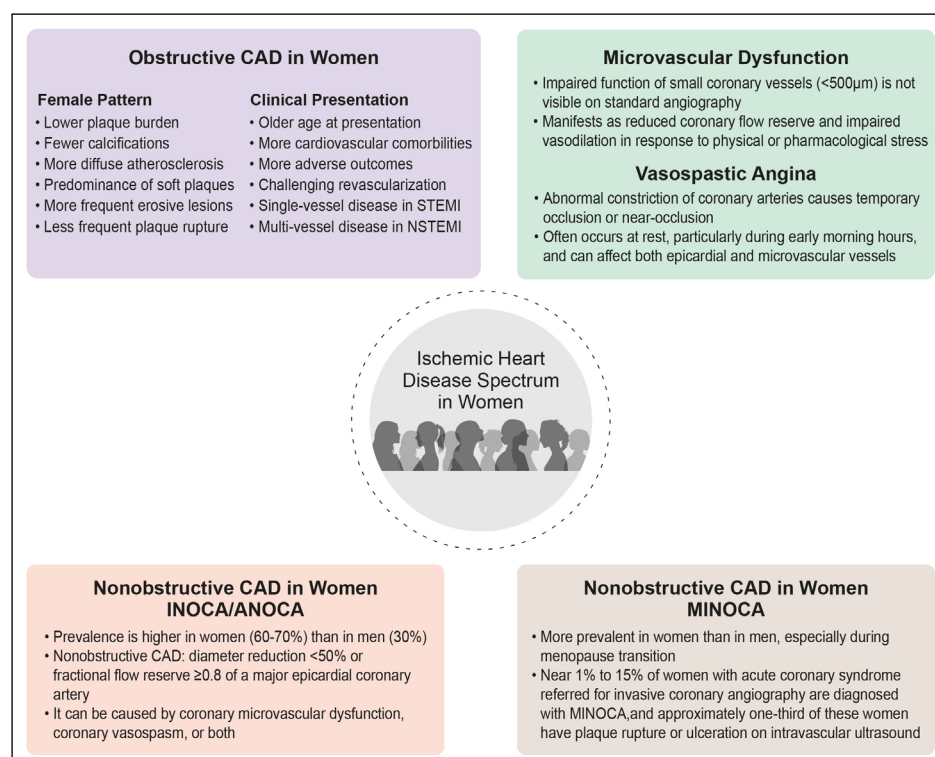


Figure 1 The spectrum of ischemic heart disease in women (22–26).

CAD = coronary artery disease; ANOCA = angina with nonobstructive CAD; INOCA = ischemia with nonobstructive CAD; MINOCA = myocardial infarction with nonobstructive CAD; NSTEMI = non-ST-elevation myocardial infarction; STEMI = ST-elevation myocardial infarction.

According to the 2021 Global Burden of Disease Study, there are significant differences in IHD between sexes concerning prevalence, death, and disability-adjusted life years (DALYs) worldwide. That study estimates IHD as the aggregate of myocardial infarction (MI), angina (stable IHD manifesting as chest pain) and ischemic cardiomyopathy (heart failure due to IHD) (27).

Globally, the IHD age-standardized prevalence rate decreased by 2.1% in men, whereas in women it increased by 4.8% from 1990 to 2021. However, the age-standardized mortality rate and the DALYs decreased in both sexes, predominantly in women (28).

During the same period (1990–2021), there were regional variations in the IHD burden according to the social development index (SDI) of the geographical regions. SDI is a tool used to measure a society's quality of life and progress by evaluating factors beyond economic output: it includes health, education, and income per capita.

The age-standardized mortality rate decreased in all SDI regions, except for the low-middle SDI region, with the most significant decline in the high SDI region. Similarly, the age-standardized DALYs decreased in all five SDI regions: the most significant decline was observed in high SDI region and the least in low-middle SDI region, especially in women (29). Contributing factors may be sex-based inequities in healthcare, ethnic minorities, socioeconomic status, along with poor awareness about CVD as a leading cause of death among women. An online survey on US women (≥ 25 years of age) showed that awareness of CVD as a major cause of death among women, declined from 65 to 44% between 2009 and 2019. This decline was more significant among Hispanic and non-Hispanic Black women and in 25-to 34-year-olds (30).

CAD usually occurs 7–10 years later in women than in men, primarily due to the protective effect of endogenous estrogens on some risk factors, including the atherosclerotic process (Table 1).

ENDOGENOUS ESTROGEN EFFECTS	MECHANISM OF ACTION
VASODILATION	↑ endothelial production of prostacyclin and nitric oxide synthase activation (31–33) ↓ endothelin synthesis (31) - Block of calcium channels (31)
IMPROVED LIPID PROFILE	↑ HDL-cholesterol efflux capacity (33, 34) ↓ LDL-cholesterol, ↓ triglyceride, ↓ lipoprotein Lp(a) (31, 35, 36)
ANTIOXIDANT EFFECTS	- Regulation of mitochondrial function, which has a pivotal role in maintaining cardiometabolic health (37, 38) ↑ sirtuin transcription through ERα, in particular SIRT3, which reduces reactive oxygen species (ROS) in the mitochondria (39, 40)
ANTI-INFLAMMATORY AND ATHEROPROTECTIVE EFFECTS	↓ inflammatory molecules and cell adhesion particles (31) ↓ vascular smooth muscle cell differentiation (41) ↓ vascular calcification (42)
ANGIOGENESIS	↑ endothelial cell migration, proliferation and survival (43) ↑ vascular endothelial growth factor (VEGF) expression in endothelial cells (43)
CARDIAC FIBROSIS PREVENTION	Prevention of cardiac fibrosis and detrimental myocardial extracellular matrix remodeling by modulating fibroblast proliferation and suppressing pro-fibrotic genes (32)
RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM (RAAS) DOWNREGULATION	↑ active form of bradykinin promoting nitric oxide (NO) production and vasodilation (31, 44)
CIRCADIAN RHYTHM REGULATION	Modulation of circadian metabolic rhythms, influencing the expression of various circadian genes through Estrogen Response Element (ERE) (45). Circadian rhythms affect cardiovascular function and therefore CVD susceptibility (46).

Table 1 Direct and indirect protective effects of endogenous estrogens on the cardiovascular system.

NO = nitric oxide; HDL = high-density lipoprotein; LDL = low-density lipoprotein; SIRT = sirtuin; ER = estrogen receptor; ROS = reactive oxygen species; VEGF = vascular endothelial growth factor; ERE = estrogen response element; CVD = cardiovascular diseases.

Anatomically, the women's epicardial coronary arteries are smaller in diameter than those of men, even when adjusted for body surface and left ventricular mass. They also have higher flow rates, which contribute to higher endothelial shear stress and the development of nonlaminar flow patterns in younger individuals (47). The functional aspects of the coronary vasculature

are influenced by endogenous estrogens, as they have a positive impact on arterial compliance due to their effects on the vascular endothelium, extracellular matrix, and smooth muscle (48).

There is a steady increase in the risk of CAD after menopause transition, which is accelerated in women who experience premature ovarian insufficiency (POI) (age at menopause <40 years) or early menopause (age at menopause between 40 and 44) (49).

The decline in estrogen levels during menopause appears to reduce nitric oxide production in endothelial cells, promoting endothelial dysfunction and release of proinflammatory cytokines, contributing to oxidative stress and endothelial injury.

Additionally, coronary vasomotor disorders, such as coronary artery spasm and/or coronary microvascular dysfunction, which are more common in women, contribute to IHD pathophysiology (22).

Obstructive epicardial CAD, characterized by the presence of atherosclerotic plaques in coronary artery walls, is the most frequent substrate of IHD in women. On clinical presentation, women with IHD are usually older than men and have an increased prevalence of traditional cardiovascular comorbidities (diabetes, hypertension, chronic kidney disease, etc.), which contribute to more adverse cardiovascular outcomes (24). Plaque rupture occurs less often in women than in men, whereas plaque erosion is more common in women (50, 51). Also, in the setting of acute myocardial infarction (AMI), culprit-artery revascularization can be more challenging in women because of the increased access-site bleeding and the smaller and more tortuous nature of their coronary arteries (Figure 1) (25).

Ischemia/angina with nonobstructive CAD (INOCA/ANOCA) occur in patients who have symptoms suggestive of ischemia, frequently positive noninvasive functional stress tests, and epicardial arteries that do not show significant obstruction. The Women's Ischemic Syndrome Evaluation (WISE) has demonstrated that patients with INOCA have a lower risk of death, MI, rehospitalization and repeat coronary angiography as compared to patients with more typical obstructive CAD (26).

The etiologies of MINOCA include coronary thrombosis, coronary embolism, coronary microvascular dysfunction, coronary spasm, myocardial bridging and plaque erosion/rupture. Relatively few women with MINOCA (14%) present with STEMI, but ST-elevation is associated with a higher risk of an adverse outcome (Figure 1) (23).

Notably, angina is more frequent in women, although they have less extensive epicardial CAD and less severe myocardial ischemia than men. In women, obstructive and nonobstructive disease often coexist with microvascular disease and vasospastic angina. The sex-related differences in symptom presentation, the lower accuracy of some diagnostic tests for obstructive and nonobstructive CAD, and less timely treatment in cases of myocardial ischemia may contribute to the unfavorable outcomes observed in women (52).

RISK FACTORS FOR CARDIOVASCULAR DISEASE IN WOMEN

TRADITIONAL RISK FACTORS

Traditional CVD risk factors are well-established, but several sex differences are involved that must be taken into account. Many of them become more prevalent during or after the perimenopausal period, changing the cardiovascular risk in menopausal women (Figure 2).

An individual woman's cardiovascular risk depends upon the number and severity of her own risk factors.

Family history

A first degree relative (parent, sibling) with early-onset (<55–60 years in men, <65 years in women) CVD increases an individual's own risk (e.g. MI, acute coronary syndrome, peripheral vascular disease, sudden cardiac death) (53–56).

Hypertension

Hypertension is the most prevalent risk factor for CVD. Although women have a slightly lower prevalence of hypertension than men in the US (44.6% vs. 50.8%), prevalence increases after

age 60 in women, becoming similar to men's at older ages (70.6% vs. 72.7%) (57). Aging increases the risk of hypertension, but there are significant sex differences in the blood pressure trajectories, particularly for the changes in systolic blood pressure. Indeed, although systolic blood pressure is lower in women than in men until about 55 years, the rate of increase in systolic blood pressure from baseline is more rapid in women than in men (58). These sex differences in blood pressure changes may explain the pathophysiological sex differences in the effect of hypertension, in addition to the sex differences in the responsiveness to antihypertensive medications. To date, there are not enough data to support sex differences in thresholds for blood pressure (59).

In women at reproductive age, the risk of hypertension may be influenced by the type and duration of hormonal contraceptive use. Blood pressure elevation has been observed in women on oral combined hormonal contraceptives (CHC) containing ethinyl estradiol at doses ranging from <30 to >50 µg, combined with 1st, 2nd and 3rd generation progestins at various doses (60). However, 4th generation CHC containing estradiol valerate and estetrol, which are natural estrogens identical to those produced endogenously, vaginal ring, as well as progestogen only pills, subdermal implant and hormonal intrauterine systems do not appear to elevate blood pressure (60–62). Moreover, CHC containing drospirenone may actually lower blood pressure through their anti-mineralcorticoid effects (63).

As to duration of use, a meta-analysis of 24 studies demonstrated that for every 5 years of 1st, 2nd and 3rd generation oral CHC use there was a 13% increased risk of hypertension (64).

Diabetes

CVD is the leading cause of morbidity and mortality in individuals with diabetes. Women with either type 1 or type 2 diabetes have a greater IHD risk than women without diabetes (65–70). The Swedish National Diabetes Registry demonstrated that women compared to men with Type 1 diabetes have higher risk of nonfatal MI or CVD mortality (HR: 3.16 vs. 1.3 for those with HgA1c <6.9%, and 10.7 and 31.8 for HgA1c >9.7). The excess risk for AMI was substantially lower in individuals with good glycemic control and the absence of renal complications, but less so in women (67).

Women with type 2 diabetes have an estimated 25–50% greater risk of CVD than men with type 2 diabetes (66, 71–73). A meta-analysis, that included 858,000 patients from 64 cohorts, also showed that women with type 2 diabetes had a 40% excess risk of IHD, compared with men (73). Despite established disparities in the management of type 2 diabetes in women compared with men, this alone does not explain the excess risk in women (66).

Dyslipidemia

According to the National Health and Nutrition Examination Survey (NHANES) data from 2017–2020, improvements have been made in the treatment of cholesterol, lowering the average cholesterol in both men and women, even if cholesterol levels remain higher in women than men (190 mg/dL vs. 183.9 mg mg/dL respectively) (74). This population average may reflect the undertreatment of elevated cholesterol in women in both primary and secondary prevention (75–77). It may also reflect the fact that women are more likely to decline statins or discontinue therapy due to perceived or experienced side effects (77, 78).

Women going through menopause have sex specific changes in cholesterol. There is a rise in low-density lipoprotein (LDL)-cholesterol (LDL-C), only partially due to ageing, as there are data supporting that some LDL-C changes are independently affected by menopausal status (79). In particular, total cholesterol, LDL-C, apo-B, high-density lipoprotein (HDL) subclasses and lipid content change markedly at the time of menopause (80). During this period, there is also an increase in lipoprotein (a) [Lp(a)] in women, in contrast with men, who have a relatively steady Lp(a) throughout adulthood (81, 82).

Obesity

Obesity is pandemic, and despite regional variations, the global age-standardized prevalence of obesity is estimated to be 18.5% in women and 14% in men in 2022, with an upward trend over time (83). Obesity is both a chronic disease and an important risk factor for many forms of CVD, including IHD. The Nurses' Health Study demonstrated that obesity was the strongest predictor

for type 2 diabetes in women (84). There is evidence that the pattern of ectopic fat in the abdomen, pericardium and neck is associated with greater CVD risk in women than men (85). Though weight gain tends to increase linearly with aging, body composition, such as central adiposity, changes significantly at menopause (86). Additionally, there are sex differences in the response to some of the newer medications used to promote weight loss, with a reported greater weight loss effect in women than men using glucagon-like peptide-1 receptor agonists (GLP-1RA) (87, 88).

Chronic Kidney Disease

Chronic kidney disease, defined as a reduced estimated glomerular filtration rate or excess urinary albumin excretion or both, is associated with an increased risk of CVD. In women, the cardiovascular-kidney-metabolic (CKM) syndrome has a lower prevalence than in men but it is associated with a higher mortality risk, as demonstrated by a recent NHANES report (89). Additionally, preeclampsia is a woman sex-specific risk factor for chronic kidney disease (90).

Systemic inflammation and autoimmune disorders

Systemic inflammation is more prevalent in women than men, as documented by higher levels of high sensitivity C-Reactive Protein (hsCRP), in addition to other biomarkers. HsCRP has been demonstrated to be a strong independent predictor of CVD risk in a number of trials (91, 92). Additionally, autoimmune diseases, which are associated with systemic inflammation, may accelerate atherosclerosis and IHD and occur more frequently in women than men. A population-based study demonstrated that individuals with autoimmune diseases have a higher risk of incident CVD, compared with age-matched controls. These events occurred more frequently in patients with systemic sclerosis, Addison's Disease, systemic lupus erythematosus or type 1 diabetes (93). A complex interaction among autoimmune disease-specific factors, persistent inflammation, accelerated atherosclerosis, microvascular dysfunction and effects of anti-inflammatory drugs, may underlie the increased CVD risk in individuals with autoimmune diseases, along with the higher prevalence of traditional CVD risk factors in this population (94).

Migraines

Two thirds of migraines occur in women. Migraines are the third most prevalent condition worldwide (95) and are associated with an increased CVD risk (96–98). The American Migraine Prevalence and Prevention study, conducted on 120,000 US households, demonstrated a strong association between migraines (with or without aura) and both CVD risk factors and CVD (99).

The exact mechanism of how migraines increase the risk of CVD remains to be clarified. Even if there is an increase in traditional CVD risk factors in migraineurs (100, 101), the endothelial dysfunction (102) and increased platelet aggregation (103, 104) associated with migraines may represent the underlying specific link between migraines and CVD.

Additionally, the chronic use of nonsteroidal anti-inflammatory drugs (NSAIDs) in some individuals with migraines could theoretically contribute to an increased CVD risk (105), although data from the Danish nationwide cohort were unable to demonstrate that the increased CVD risk was due to NSAID use (106).

Sleep disorders

Inadequate sleep is associated with CVD and its risk factors (107, 108). Over one third of adults report insufficient sleep (<7 hours), with more women reporting it than men (109, 110). Women are at a particularly high risk for nonoptimal sleep during the perimenopause and post-menopause phases, when hormone fluctuations may lead to vasomotor symptoms (VMS) that can disrupt sleep. Indeed, there is an association between VMS severity, night sweats and self-reported poor sleep quality (111). Poor sleep quality has also been associated with increased inflammation biomarkers, such as IL-6, CRP and fibrinogen, in women but not in men (112).

Sleep apnea, which is known to disrupt sleep and to negatively affect its quantity and quality, is associated with an increased CVD risk (113). Sleep apnea is more prevalent in men than women. Women usually have shorter apneic episodes and lower apnea-hypopnea index scores than men, but experience more prolonged partial upper airway obstruction. However, when they experience long apneic episodes, women have a more severe oxygen desaturation

than men. Moreover, women with sleep apnea experience more insomnia and are often more symptomatic than men (114). Women with untreated sleep apnea, have a 3.5 times greater risk of dying from CVD compared with women without sleep apnea (113, 115).

Tobacco use

Cigarette smoking is the leading preventable cause of cardiovascular deaths. Although the prevalence of smoking is lower in women compared with men, cigarette use may be more detrimental in women than men (116). Even infrequent smoking, which is more common among women, has a stronger association with IHD, particularly STEMI, in women than men (117). Additionally, the cardiovascular impact of dual use of smoking and combined estrogen-progestin oral contraceptives, especially after 40 years, is associated with an increased risk of AMI as compared to smoking alone, which has been attributed to the combination of their pro-thrombotic effects (118).

Low physical activity

Physical activity is inversely associated with IHD risk. A recent study on over 400,000 US adults has demonstrated that women derived greater gains in all-cause and cardiovascular mortality risk reduction than men from the same amount of physical activity for both aerobic and resistance training (119). However, women are less likely to meet physical activity guideline recommendations than men (20% vs. 28%), and this worsens with age (120). Nonetheless, physical activity measurements do not account for domestic activities such as, cooking, cleaning and childcare, hence current tools to assess physical activity often underestimate actual activity in women.

SEX-SPECIFIC RISK FACTORS (FIGURE 2)

Adverse Pregnancy Outcomes

Gestational hypertension: eclampsia, preeclampsia and pregnancy-associated hypertension

Gestational hypertension of any sort is associated with an increased risk of developing hypertension, chronic kidney disease, diabetes, stroke [OR: 1.83 (95% CI: 1.79–4.22)] and cardiovascular disease (including heart failure and MI) [OR: 1.67 (95% CI: 1.28–2.19)] later in life (121, 122). The risk increases with increasing severity of disease from pregnancy associated hypertension to preeclampsia and eclampsia. A prospective, observational UK Biobank study on more than 220,000 women between 40 and 69 years (mean age at baseline 57.4 yrs), who reported ≥ 1 live births and were followed up for a median of seven years, showed that those who had any hypertension during pregnancy were at greater risk of coronary artery disease [HR: 1.8 (95% CI: 1.3–2.6)], heart failure [HR: 1.7 (95% CI: 1.04–2.6)], aortic stenosis [HR: 2.9 (95% CI: 1.5–5.4)] and mitral regurgitation [HR: 5.0 (95% CI: 1.5–17.1)] (123). Data from the French National hospitalization database demonstrated that women with gestational hypertension had a five-fold greater risk of cardiovascular death (HR: 5.46, 95% CI: 1.93–15.49) (124).

Gestational diabetes

Gestational diabetes is strongly associated with an increased risk of CVD (125). In a pooled analysis of 9 studies on over 5 million women, those with gestational diabetes had a two-fold risk of cardiovascular events in the first 10 years postpartum, compared with women without gestational diabetes, even when they did not develop type 2 diabetes in that time period (126). Moreover, women with both gestational hypertension and gestational diabetes had an increased risk of MI (HR: 23.33, 95% CI: 4.84–112.39), within the first 5 years post-partum (124).

Preterm delivery

Preterm delivery, defined as a delivery prior to 37 weeks' gestation, complicates about 11% of pregnancies worldwide and is strongly associated with maternal risk of CVD and stroke. The greatest risk is when deliveries occurred prior to 32 weeks' gestation (127).

Small-for-gestational-age offspring

Having had a small-for-gestational-age offspring is associated with an increased maternal risk of CVD. This association has been demonstrated to be dependent upon the quartiles of the offspring birth weight in first and second pregnancies: women with a first offspring in the lowest quartile are at an increased CVD mortality risk, in both iatrogenic and spontaneous

deliveries. Nonetheless, in women with a second offspring in the highest quartile, the increased risk appears to be eliminated (128).

It is important to note that there may be an overlap with hypertensive pregnancy disorders, gestational diabetes, placental abruption, preterm delivery, assisted reproductive technologies and smoking habits, all of which are frequently associated with the delivery of small-for-gestational-age offspring.

Miscarriage/stillbirth

Pregnancy loss is associated with an increased CVD risk. A large meta-analysis demonstrated that women with recurrent miscarriages and/or one or more stillbirth had an increased risk of CVD (129). Additionally, the WHI demonstrated that in post-menopausal women with a history of a stillbirth, there was an increased risk of CVD, heart failure and stroke (130).

Assisted Reproductive Technologies

Fertility hormonal therapies are used widely across the globe. Currently, the use of fertility therapy is not considered an independent risk factor for ASCVD (131). However, there are data to suggest that women who have failed fertility therapy have an increased risk of future ASCVD events (132). It is difficult to draw firm conclusions about any of the assisted reproductive therapies (ART). To date, most ART recipients have higher socioeconomic status, compounded with better access to care, including health insurance, and other confounders that limit extrapolation. Additionally, women undergoing ART often have more pre-existing CVD risk factors, such as obesity, diabetes, hypertension and polycystic ovarian syndrome (PCOS) (133). Further studies are needed to clarify the extent to which any observed CVD risk is attributable to ART itself rather than underlying patient characteristics.

Age at Menarche

Early or late onset of menarche is associated with an increased risk of CVD including MI, stroke and heart failure hospitalizations. Indeed, the WISE study showed that women who had menarche at ≤ 10 years or at ≥ 15 years had a higher risk of CV events [HR: 4.53 (95% CI: 2.13–9.63) and 2.58 (95% CI: 1.28–5.21), respectively], compared with reference women with menarche at 12 years (134).

Polycystic ovarian syndrome

Polycystic ovarian syndrome (PCOS) is associated with an increased risk of impaired glucose tolerance, metabolic syndrome and diabetes compared to women without PCOS (135). Nonetheless, the data are unclear as to whether PCOS is an independent risk factor for premature CVD in women (i.e. < 65 yrs). A Danish registry study on women undergoing assisted reproductive therapies demonstrated an increased CVD risk in those with PCOS, compared with women without PCOS (136). In contrast, the WISE study on postmenopausal women with suspected myocardial ischemia did not demonstrate any association with PCOS and CVD (135).

Functional hypothalamic amenorrhea

Functional hypothalamic amenorrhea is one of the causes of ovarian dysfunction and can occur due to psychological stressors or metabolic insults, such as caloric restriction or excessive exercise. Endothelial dysfunction and premature CVD have been associated with functional hypothalamic amenorrhea (137, 138).

Premature ovarian insufficiency and early menopause

Both age at menopause and type of menopause are related to CVD risk. Premature ovarian insufficiency (POI) leading to premature menopause (onset < 40 years) and early menopause (onset between 40 and 44), coupled with an extended period of estrogen deprivation, are associated with a greater risk of CVD (139, 140). A meta-analysis, that included over 190,588 women, demonstrated that those with POI had an increased incidence of CVD (141). Another meta-analysis on 310,329 women reported that those who experienced premature or early-onset menopause were at higher risk of both total and lethal IHD, than women who reached menopause at ages ≥ 45 years, even after adjusting for potentially confounding cardiovascular risk factors (142). A UK Biobank study also demonstrated that the earlier that natural menopause occurred, the greater the risk of IHD and stroke (143).

Moreover, surgical menopause has been associated with a higher cardiovascular risk, as women who undergo bilateral oophorectomy without appropriate supplemental hormonal therapy, face an increased CVD risk and tend to have more arterial stiffness than those who experience natural menopause (144, 145).

Endometriosis

Endometriosis is a complex chronic, multisystemic inflammatory condition affecting one in ten reproductive-aged women (146). The better understanding of the underlying pathophysiologic mechanisms of both endometriosis and CVD has prompted scientific interest on the interaction between the two (147). A recent meta-analysis on case-control and cohort studies demonstrated that women with endometriosis had a 23% higher risk of developing CVD than those without. Possible reasons for this association include shared underlying chronic systemic inflammation, increased oxidative stress, endothelial dysfunction, increased cellular proliferation and atherogenic lipid profiles. Common genetic and lifestyle risk factors (e.g. physical inactivity, unhealthy diet, alcohol consumption and tobacco use) may contribute to the link between endometriosis and CVD. However, due to the limitations in the number and heterogeneity of the studies included in the meta-analysis, no definitive causal relationship can be established to date (148).

Breast Cancer and Related Therapies

Breast cancer is the most common cancer in women globally. Survival from breast cancer has significantly increased due to improved early detection and advancements in therapies. Breast cancer itself is associated with a number of overlapping risk factors for cardiovascular disease, including physical inactivity, poor diet, tobacco use, diabetes and obesity (149). Additionally, both chemotherapy and radiation therapy have cardiotoxic effects and are associated with an increased risk of heart failure and IHD (149–151).

Moreover, breast arterial calcification (BAC), which is incidentally visible on screening mammograms, has been found to be independently associated with all-cause mortality and CVD, especially in women under the age of 60 years (152–154). Beyond simple presence/absence, BAC measurement may provide incremental risk stratification. In the future, quantification of BAC using artificial intelligence-based algorithms may become clinically useful for CVD risk assessment (152).

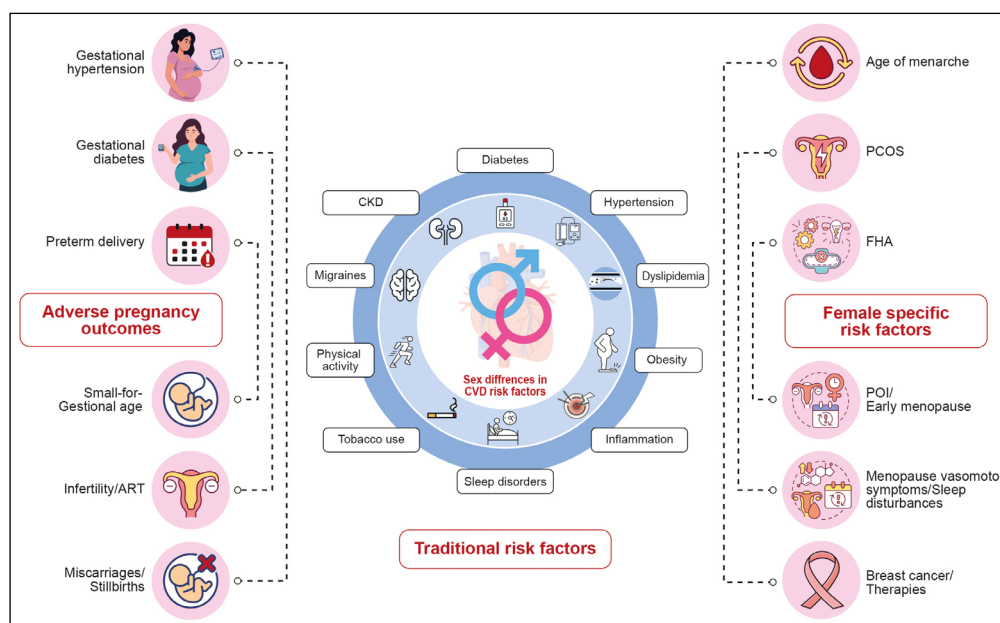


Figure 2 Risk Factors of CVD in Women.

Risk factors for CVD in women include sex differences in traditional risk factors, adverse pregnancy outcomes that are specific to women and female specific risk factors.

ART = artificial reproductive therapies; FHA = functional hypothalamic amenorrhea; PCOS = polycystic ovarian syndrome; POI = premature ovarian insufficiency; CKD = chronic kidney disease.

MENOPAUSE

DEFINITIONS

Menopause is a natural process defined as the permanent cessation of menstruation, due to the irreversible loss of ovarian follicular activity, leading to a significant decline in estrogen and progesterone production. It marks the end of a woman's reproductive function and is clinically

diagnosed retrospectively after twelve consecutive months of amenorrhea, in the absence of other causes. The decline in sex steroid levels leads to multiple changes in physiological functions and typically occurs over a period of several years (155).

Considering the final menstrual period (FMP) as point 0, the following terminologies are recommended to define the different stages of the reproductive aging process (Table 2):

- Menopausal transition, which is divided into early menopausal transition (Stage -2), characterized by menstrual irregularities that start when menstrual cycle lengths vary by 7 days or more between consecutive cycles and late menopausal transition (Stage -1), defined by a period of amenorrhea lasting 60 days or more, in addition to an increased follicle-stimulating hormone (FSH) level >25 IU/L (156, 157).
- Post-menopause, which is divided into early and late post-menopause. Early post-menopause is further subdivided into three stages to account for the continuous increase in FSH and decrease in estradiol for 2 years after the FMP. Stage +1a corresponds to the 12 months post FMP. Stage +1b lasts one year and includes the remaining period of rapid changes in FSH and estradiol levels. Stage +1c, lasting for about 3–6 years, represents the period of stabilization of high FSH and low estradiol levels. Late post-menopause (Stage +2) lasts for the rest of the lifespan and is characterized by greater hormonal stability (156–158).

The term perimenopause is still widely used to describe the time around menopause and includes the early and late menopausal transition (Stages -2 and -1) as well as the first year after the FMP, i.e. the early post menopause Stage +1a (156, 157).

The term climacteric can be used to describe the period of the gradual decline in ovarian function characterized in most women by the onset of signs and symptoms related to estrogen deprivation (159).

Menopause is classified as:

- *Premature* when FMP occurs before the age of 40
- *Early* when FMP occurs between the ages of 40 and 44
- *Late* when FMP occurs after the age of 55

FMP (0)	TERMINOLOGY (STRAW Stage)	DURATION	ESTRADIOL Serum levels
	Early MENOPAUSAL TRANSITION (-2)	PERIMENOPAUSE	Variable
	Late MENOPAUSAL TRANSITION (-1)		~ 60 pg/mL
	Early POSTMENOPAUSE (+1a)	1-3 Years	~ 55 pg/mL
	Early POSTMENOPAUSE (+1b)	1 Year	~ 30 pg/mL
	Early POSTMENOPAUSE (+1c)	1 Year	≤ 20 pg/mL
	Late POSTMENOPAUSE (+2)	3-6 Years	≤ 20 pg/mL
		Remaining lifespan	≤ 20 pg/mL

Table 2 Terminology, Stages of Reproductive Aging (STRAW+10), duration and mean levels of estradiol during the menopausal transition and post-menopause (156–158). FMP = final menstrual period.

EPIDEMIOLOGY

A comprehensive systematic review of 46 studies carried out in 24 countries reported a global mean natural menopause age of 48.8 years (160). However, geographical differences do play a significant role in the age of natural menopause. African, Latin American and Asian populations tend to experience a significant earlier onset, while European, North American and Australian women have a higher mean age at natural menopause (160, 161). Socioeconomic position and lifestyle factors seem to be involved in such differences (160), together with dietary factors which appear to contribute to either an earlier or later onset (162, 163).

Smoking is consistently associated with earlier menopause across diverse populations (164), whereas higher educational attainment, prior oral contraceptive use and higher body weight are all associated with delayed menopause (165). Importantly, there are increasing data suggesting a bidirectional causality between the timing of menopause and CVD risk. On the one hand, premature or early menopause are CVD risk factors; on the other, women with poor cardiovascular health have an increased risk of an earlier onset of natural menopause. This was also demonstrated by the Framingham Heart Study, in which premenopausal women with a worse cardiovascular risk profile, particularly elevated cholesterol, and hypertension, were more likely to have early menopause (140, 166–168).

Table 3 summarizes the main factors known to be associated with earlier or delayed menopause onset.

Natural menopause occurs in most women between 45 and 55 years of age. However, 3.7% of women experience POI, which results in a premature menopause (169); early menopause has a global prevalence of 12.2% (170). Underlying causes of non-iatrogenic POI involve idiopathic reasons, which account for 85–90% of the cases, including environmental factors, genetic, metabolic and/or autoimmune diseases (171). Iatrogenic causes include radiotherapy, chemotherapy and surgical procedures on the ovaries, all of which are on the increase (172, 173). At the other end of the spectrum, late menopause is experienced by about 7% of women (169, 170).

CATEGORY	FACTORS ASSOCIATED WITH EARLIER MENOPAUSE ONSET	FACTORS ASSOCIATED WITH DELAYED MENOPAUSE ONSET
Lifestyle	- Cigarette smoking - Too vigorous physical exercise - Low lifelong sun exposure	- Moderate alcohol consumption - Regular tea consumption - Moderate physical activity
Dietary	- Vegetarian diet - High polyunsaturated fat intake - Caloric restriction particularly in childhood	- High total caloric intake - High fruit and vegetables intake - High protein and carbohydrate intake
Socioeconomic	- Lower education level - Unemployment	- Higher educational attainment - Being employed
Reproductive History	- Nulliparity	- Parity
Genetic/Demographic	- BRCA mutations	- Japanese ethnicity
Other	- History of heart disease - Cardiac risk factors - Increased epigenetic age	- Higher BMI in early adulthood - Prior oral contraceptive use

Table 3 Factors associated with earlier or delayed menopause onset (162–166, 174).

BRCA = BRest Ancer gene.

EARLY AND LONG-TERM EFFECTS OF MENOPAUSE ALONG WITH AGING AND THEIR IMPACT ON WOMEN'S HEALTH

The hormonal shifts caused by the gradual failure in ovarian function, driven by the depletion of gonadotropin-responsive follicles, are associated with a wide range of early signs and symptoms of varying duration and severity as well as long-term effects that are further influenced and amplified by the natural aging process (175, 176) (Figure 3). For example, some of the changes in cardiovascular risk factors associated with menopause are mainly related to the estrogen drop (lipid profile, metabolic syndrome and central adiposity), while others are predominantly related to chronological aging (weight, blood sugar, insulin resistance and hypertension), all of which contribute to an increased cardiovascular risk profile (140).

All signs and symptoms typical of menopause may impact various aspects of a woman's health and well-being and can be very distressing (177). In addition, more than 80% of women report multiple menopausal symptoms (178).

According to a systematic review and meta-analysis of 482,067 women, globally, the most prevalent reported menopause symptoms are joint and muscular discomfort, physical and mental exhaustion, irritability, poor memory, hot flashes, sleep problems, anxiety, mood swings and sexual problems (179).

Central nervous system symptoms and disorders may be due to the neurobiochemical changes and adaptations, following ovarian failure (177). Persistent vasomotor symptoms, in particular hot flashes and night sweats, affect 75% of women during menopause (177) and are considered severe in 25% of cases (180). Symptoms may last for years, in 39% of women aged 65–69, 31% of those aged 70–74, and 24% of women between 75 and 79 (10, 181). Vasomotor symptoms are the result of disrupted thermoregulatory control in the hypothalamus, primarily driven by a decline in estrogen levels (180, 182) and have been reported more frequently by African women, with a 64% prevalence, followed by Asian (51%), European (54%) and North American women (52%) who have a comparable prevalence, whilst Oceanian women have the lowest prevalence (40%) (179, 180). These symptoms tend to occur suddenly and unpredictably, making them some of the most bothersome complaints among menopausal women (179). Other commonly reported central nervous system symptoms include physical and mental exhaustion (over 60% of women) (179), sleep disturbances (35–60%) (183), irritability, anxiety and mood swings (about 50%) and depression (43%) (179, 184). There may also be cognitive effects (185), which may range from mild memory lapses to more pronounced attention and memory difficulties, often accompanied by a sense of mental confusion—collectively referred to as ‘brain fog’—up to cognitive impairment (186). Postmenopausal women are at an increased risk of developing dementia, which may be related to the reduced estrogen neuroprotective effects, as well as increase in systemic hypertension (187). Clinically, all these symptoms may negatively affect a woman’s quality of life, even interfering with her daily activities and work performance (188).

The hypoestrogenic state, together with the progressive decline in androgen levels associated with ovarian and adrenal aging (189), play pivotal roles in the development of sexual dysfunctions and genitourinary syndrome (GSM), with a prevalence of 40–54% in postmenopausal women (190). Symptoms include recurrent lower genital tract infections, vaginal dryness, increased vaginal pH, dysuria, reduced libido, and female sexual interest/arousal disorder (177, 191). These symptoms rarely improve without adequate intervention. If ignored or poorly managed, they can therefore have a negative impact on a woman’s self-esteem and/or intimate relationships (192).

Musculoskeletal discomfort is one of the most common complaints reported by menopausal women, with an estimated overall prevalence of 71% (193) and is often associated with a decline in muscle mass (194). The loss of muscle mass, strength and function is generally defined as sarcopenia (195).

Moreover, osteoporosis and osteopenia, that have an estimated prevalence of 27% and 42% in postmenopausal women, respectively (196), are worsened by the reduction in endogenous estrogens, which normally play a crucial role in regulating bone remodeling. When there is an estrogen deficiency, resorption outweighs formation, which can lead to loss of bone mass and increased fracture risk, severely impacting mobility and the quality of life (197, 198).

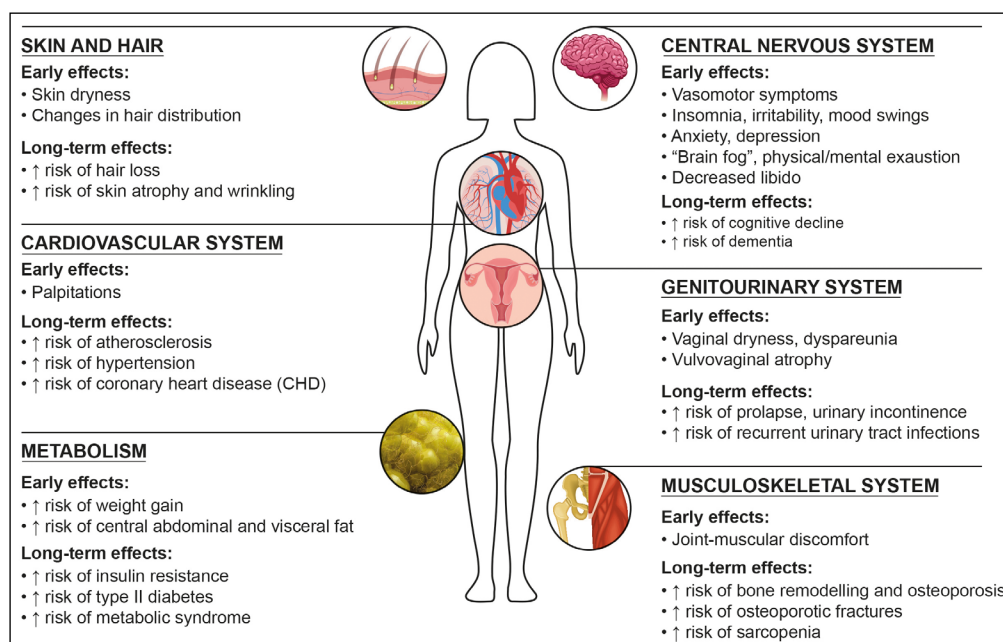


Figure 3 Early and long-term effects of menopause along with aging on body systems and organs (183–187, 190–195, 198–200, 206, 210, 212, 215).

Endogenous estrogens exert several beneficial effects on a women's cardiovascular health, including through their protective role on the endothelial function (44) (Table 1). Dysregulation of endothelial function may contribute to and mediate cardiovascular disease (CVD) (31, 201). The lower CVD incidence observed in premenopausal women compared to men in the same age group suggests an influence of endogenous sex hormones in modulating cardiovascular risk (202) and this advantage declines after menopause onset (203). Postmenopausal women who experienced late menopause have significantly better endothelial function than those with earlier menopause (201). These vasoprotective effects are mediated through both genomic and non-genomic pathways, involving the estrogen receptors ER- α and ER- β (204).

Menopause has also been associated with the development of subclinical atherosclerosis (144), as measured by carotid intima-media thickness, but not by other tests such as coronary artery calcium (9). In one study on 110 postmenopausal women, those with earlier menopause (before age 45 years) had greater atherosclerotic plaque volume, assessed by ultrasound in peripheral arteries, and a more rapid disease progression (205).

In addition to endothelial dysfunction, postmenopausal estrogen deficiency is associated with increased vascular resistance which may be due to structural changes in the arterial walls, including collagen deposition and reduced elastin, which contributes to arterial stiffness. There is also an upregulation of the Renin-Angiotensin-Aldosterone System (RAAS), resulting in increased angiotensin II production, a potent vasoconstrictor, and aldosterone which promotes sodium retention and rise in blood pressure (206). Hypertension, in particular isolated systolic hypertension (206), has a two-fold higher prevalence in postmenopausal women, especially in earlier onset of menopause (<45 yrs), compared to premenopausal women (207, 208).

Menopause and chronological aging are both associated with significant alterations in metabolic biomarkers, including higher total cholesterol, triglyceride, LDL cholesterol and lipoprotein Lp(a) levels, along with reduced HDL cholesterol (144). Together, these changes contribute to a more atherogenic lipid profile and increased cardiovascular risk. Estrogen decline is the principal mechanistic driver of these changes. Loss of estrogen increases hepatic proprotein convertase subtilisin/kexin type 9 (PCSK9) expression, accelerating LDL-receptor degradation and elevating circulating LDL cholesterol. It also diminishes cholesterol efflux from vascular smooth muscle cells and enhances hepatic and endothelial lipase activity, promoting the conversion of large, cardioprotective HDL particles into smaller, less functional forms. Furthermore, estrogen normally suppresses hepatic apo(a) synthesis; its decline therefore leads to increased Lp(a) concentrations and enhanced atherothrombotic potential (36). Consistent with these mechanisms, postmenopausal women receiving MHT consistently show lower Lp(a) levels than untreated women (209).

Furthermore, postmenopausal women have a decreased basal metabolism, attributed to a complex interplay of endocrine and metabolic factors, including a reduction in lean body mass, which contributes to weight gain (210, 211). In addition, they exhibit an increased central abdominal and visceral fat distribution (212), related to the absolute estrogen deficiency and relative androgen excess, which is typical of menopause (213, 214).

Also, impaired glucose metabolism and insulin resistance, potentially worsened by estrogen deficiency (215, 216), increase the risk of metabolic syndrome and type II diabetes, with higher risk of CVD and mortality (144, 217).

MENOPAUSE HORMONE THERAPY (MHT)

INDICATIONS AND EFFECTS OF MHT

MHT is the first-line treatment to manage menopause-related symptoms (218) to help improve the overall quality of life for women during the menopausal transition and postmenopause (219). MHT may provide potential long-term health benefits, when used appropriately, tailored to the woman, taking into account her medical history, age, individual risk factors, lifestyle and comorbidities. To obtain maximum efficacy and reduce potential risks, systemic MHT should be started as early as possible after the onset of menopausal symptoms, at most within 10 years after FMP, and continued as long as the regular reassessment of the benefit and risk ratio is favorable (10, 49, 220).

The main early and long-term potential benefits and risks associated with systemic MHT are reported in [Tables 4](#) and [5](#), whereas [Table 6](#) summarizes its reported main neutral effects.

Though MHT is not currently recommended for primary or secondary prevention of CVD alone, there is newer data on relative safety and potential targeted benefits. A meta-analysis of 19 randomized controlled trials, including over 40,000 post-menopausal women who were randomly assigned to receive either oral MHT or placebo, did not find any protective effect of hormone therapies against death from any cause or from cardiovascular disease, whether in healthy women or those with pre-existing heart disease. However, in most studies the mean age of participants was over 60 years. To test the timing hypothesis, trials were therefore stratified based on when treatment was started: either <10 years (in over 9,000 women, 74% of whom from the largest WHI trials) or ≥ 10 years after menopause. They found lower all-cause mortality (RR: 0.7; 95% CI: 0.52–0.95) and a lower composite of CVD death and non-fatal MI (RR: 0.5; 95% CI: 0.29–0.96) in women who initiated treatment within 10 years after menopause or before the age of 60, compared with those who received placebo. In contrast, women who started MHT more than 10 years after menopause or when they were older than 60 years did not experience any benefit ([221](#)). In addition, a post hoc analysis of health outcomes during the intervention and extended post-stopping phases of the WHI trials, which enrolled women aged 50–79 years, showed that all-cause mortality was lower in women who started MHT between the ages of 50 and 59, compared with women on placebo and with those who started MHT at ages 60–69 or 70–79 ([222](#)).

Furthermore, a recent secondary analysis of the WHI trials—including more than 27,000 postmenopausal women aged 50–79 years—evaluated cardiovascular outcomes and vasomotor symptom relief stratified by age and symptom severity. The reassessment noted that, in women randomized at age 50–59 years, conjugated equine estrogens (CEE) alone reduced the risk of ASCVD, whereas CEE plus medroxyprogesterone acetate (MPA) did not alter risk compared with placebo. In contrast, women who initiated CEE alone at older ages (>60 years) exhibited an increased ASCVD risk. These findings further demonstrate the importance of timing when initiating MHT ([223](#)).

Table 4 Potential benefits associated with systemic MHT.
CEE = conjugated equine estrogens; MPA = medroxyprogesterone acetate.

SYSTEMIC MHT: POTENTIAL BENEFITS
• ↓ vasomotor symptoms (219 , 224)
• ↑ sleep quality , especially in patients suffering from concomitant vasomotor symptoms (225)
• ↓ depressive symptoms , when MHT is used alone or in combination with antidepressants (226 , 227)
• ↓ genitourinary symptoms (228)
• ↓ urinary tract infections (229)
• ↑ sexual function (10 , 230 , 231)
• ↓ biological aging (174 , 232)
• ↓ all-cause mortality when MHT is started <10 years after menopause or in women <60 years old (221 , 222 , 233–235)
• ↓ composite of death from cardiovascular causes and non-fatal myocardial infarction when MHT is started <10 years after menopause or in women <60 years old (221)
• ↑ high density lipoprotein (10)
• ↓ insulin resistance and the incidence of diabetes (10)
• ↓ osteoporosis, bone loss and fracture risk at all sites (236)
• ↓ sarcopenia risk in physically active women (237)
• ↓ colorectal cancer risk , especially in women with a high polygenic risk score (238 , 239)
• ↓ endometrial cancer risk with continuous use of combined estrogen-progestin, particularly in obese women (240 , 241)
• ↓ breast cancer risk with estrogen-alone therapy, especially with CEE (242)
• ↑ cognitive performance when MHT, with the exception of CEE-MPA (243), is started in the peri-menopausal or early post-menopausal period (244 , 245)
• ↓ Alzheimer's Disease risk , particularly in APOE4 carriers (246) when MHT is started <10 years after menopause (247)

SYSTEMIC MHT: POTENTIAL RISKS

- **↑ VTE risk only with specific types of MHT** i.e. oral estrogen alone or combined; transdermal estradiol combined with norepregnane, pregnane or nortestosterone derivatives (248–250) whenever it is started (221)
- **↑ ischemic stroke risk with specific types of MHT**, i.e. oral estradiol, especially at high doses; CEE, especially at intermediate-high doses or when initiated >10 years after menopause; norepregnane derivatives; tibolone (221, 222, 251–254)
- **↑ hemorrhagic stroke risk with specific types of MHT**, i.e. tibolone (253)
- **↑ urinary incontinence and pre-existing pelvic organ prolapse** (255, 256)
- **↑ breast cancer risk with specific types of MHT**, i.e. estrogens combined with levonorgestrel/norethisterone acetate/medroxyprogesterone acetate, especially for long-term (257) or tibolone if taken for >5 years (10)
- **↑ endometrial cancer risk** if systemic estrogen-alone therapy is given to women with an intact uterus (258)
- **↑ serous and endometrioid ovarian cancer risk with specific types of MHT**, i.e. estrogen-alone or estrogen-progestogen preparations (259, 260)
- **↑ serous and all epithelial ovarian cancer risk** with tibolone used for >10 years (261)
- **↓ cognitive function** when MHT is started in older postmenopausal women (243)

SYSTEMIC MHT: NEUTRAL EFFECTS

- **= VTE risk** with transdermal estradiol alone or transdermal estradiol + progesterone (250)
- **= or even ↓ ischemic stroke risk** with transdermal estradiol (251, 253, 254)
- **= hemorrhagic stroke risk** with transdermal estradiol (253)
- **= or even ↓ hemorrhagic stroke risk** with oral estradiol and CEE (253, 262)
- **= breast cancer risk** with estradiol + progesterone or dydrogesterone (263)
- **= mammographic breast density and breast cancer incidence risk with BZA/CE** (264)
- **= breast cancer risk** with Tibolone used for <5 years (263)
- **= survival from overall breast cancer** with MHT (265)
- **= triglyceride levels** with transdermal MHT, including estrogen alone and estrogen + progestogen (266)
- **= mucinous and clear cell ovarian cancers** with estrogen only and estrogen-progestogen therapy (259, 260)
- **= lung cancer incidence risk and survival** with MHT (220)

TYPES OF MHT

Currently, numerous types of MHT formulations, dosing and administration routes are available that allow for the tailoring of replacement therapy to individual needs, based on patient preference, uterine presence/absence, symptom severity, medical history and comorbidities (218, 220).

The lowest effective dose of hormone therapy should always be preferred. MHT should be started as soon as possible after the onset of menopausal symptoms and continued until benefits consistently outweigh risks. Regular reassessment of the woman and shared decision making with the patient are of paramount importance. Discontinuation based solely on age is not supported by clinical evidence. Appropriate formulation, dosing and route of administration are fundamental particularly for women with cardiovascular risk or prolonged use (10, 219, 220, 267).

Systemic MHT

- Estrogen monotherapy, oral or transdermal, is suitable for women who have had a hysterectomy. Estradiol (E2: high dose 2 mg, low dose 1 mg), 17-beta Estradiol (17βE2: patch high dose 50 µg, low dose 25 µg; gel/spray high dose 2 mg, low dose 0.5–1 mg) or Conjugated Equine Estrogens (CEE: high dose 1.25–0.625 mg, low dose 0.45–0.3 mg) can be used (10).

Oral and transdermal regimens have comparable efficacy for symptom relief and are both appropriate options (268). However, in terms of thrombotic profile, the oral route increases the risk of venous thromboembolism (VTE), whereas according to the results of case-control studies the transdermal route may not be associated with an increased VTE risk (248, 249, 269).

Table 5 Potential risks associated with systemic MHT.

VTE = venous thromboembolism; CEE = conjugated equine estrogens.

Table 6 Neutral effects of systemic MHT.

VTE = venous thromboembolism; CEE = conjugated equine estrogens; BZA/CE = Bazedoxifene/Conjugated Estrogens.

The differential impact of oral versus transdermal estrogen on VTE risk is supported by biological data: Oral estrogen therapy, unlike transdermal administration, has been shown to activate the coagulation cascade and to induce resistance to activated protein C. These prothrombotic effects are likely attributable to the hepatic first-pass metabolism associated with oral administration (270).

Evidence from some studies indicates that transdermal MHT does not raise the VTE risk even in women with known risk factors (271) such as previous VTE (272, 273), obesity (249, 270, 274), prothrombotic genetic polymorphisms (275, 276), or genetic polymorphisms that interact with MHT (e.g. cytochromes P450 3A5*1 allele) (277).

- Combined estrogen-progestogen therapy is recommended for women with an intact uterus to prevent endometrial hyperplasia and endometrial cancer, as progestogens trigger secretory transformation of the endometrium, counterbalancing estrogen-induced proliferation (278). Progestogens used in MHT include natural progesterone and synthetic progestogens or progestins, which are further divided on the basis of their chemical structure, into progesterone derivatives (dydrogesterone, medroxyprogesterone acetate, chlormadinone acetate, cyproterone acetate, megestrol acetate and norgestrel acetate), 19-nortestosterone derivatives (norethisterone and norethisterone acetate, levonorgestrel, norgestrel, gestodene, desogestrel, norgestimate and dienogest) and a spironolactone derivative (drospirenone) (279, 280).

The type of progestogen used in the combined treatment can significantly influence the risk of VTE (281). Indeed, micronized progesterone and dydrogesterone, which can both be administered either orally or vaginally, are not associated with an increased risk of VTE. In contrast, other progestins, namely MPA, norpregnane derivatives such as norgestrel acetate are linked to a significantly higher risk of VTE (250, 278, 281, 282). The effect of nortestosterone derivatives, such as norethisterone acetate (NETA) remains uncertain (281). Based on current evidence, transdermal estrogen combined with micronized progesterone is therefore likely the safest option in terms of VTE risk (250).

The MHT regimens can be sequential or continuous combined.

The sequential regimen involves daily estrogen with progestogen added cyclically (e.g. 12–14 days per month), leading to regular withdrawal bleeds, whereas, the continuous regimen uses daily estrogen and progestogen without a break (283).

Sequential MHT may be initiated during the premenopausal period, but it can also be started and/or continued after menopause, when the patient prefers to have regular withdrawal bleeding (284).

- Tibolone (high dose 2.5 mg, low dose 1.25 mg) is a synthetic steroid with estrogenic, progestogenic and androgenic properties, which can be used orally as monotherapy, regardless of uterine presence.
It has beneficial effects on mood, sexual well-being and GSM as well as vasomotor symptoms (279, 285). Tibolone is the most effective therapy in terms of restoration of desire, sexual interest and satisfaction (10). The use of tibolone is not associated with an increased VTE risk, however it has been reported to increase the risk of IHD and stroke (252, 253, 285, 286).
- Tissue-selective estrogen complex (TSEC) is a therapeutic option that combines conjugated estrogens (CE: 0.45 mg) with a newer generation selective estrogen receptor modulator (SERM), i.e. bazedoxifene (BZA: 20 mg), providing effective relief from vasomotor symptoms (287), while offering endometrial protection (288). The occurrence of VTE is low and does not appear to exceed that observed with CE alone (289).

Vaginal MHT

Vaginal estrogen preparations (estradiol, estriol, promestriene), at low doses, are available in different formulations (creams, gels, rings, tablets, suppositories and capsules) and are specifically indicated for the treatment of vulvovaginal and related sexual and urinary signs and symptoms (10, 224, 290). Vaginal preparations can also be offered to patients already on systemic treatment if indicated (291). They have a favorable safety profile, including cardiovascular events (290). Indeed, as their systemic absorption is minimal, the classical contraindications to systemic MHT do not apply to them (10).

Prasterone, a synthetic form of dehydroepiandrosterone, is also indicated for the treatment of vulvovaginal atrophy. Noteworthy, like the vaginal estrogen preparation it does not raise systemic sex steroid levels beyond the normal postmenopausal range and is generally well tolerated (292).

Due to the lack of systemic effects, both low-dose vaginal estrogens and prasterone may be considered, off-label, for the management of persistent genitourinary menopausal symptoms in women with breast cancer, after shared decision making of the potential benefits to quality of life versus risk (293).

MHT IN PATIENTS WITH CVD RISK FACTORS

The currently used risk charts, like the Systematic COronary Risk Evaluation 2 (SCORE2), may underestimate the actual risk in women at middle-age (294). Some traditional risk factors, such as smoking and diabetes, have a greater impact in women than in men. In addition, women have sex-specific risk factors that are important risk enhancers earlier in life, but their relevance diminishes after age 65, when traditional CVD risk factors (lipids, blood pressure, insulin resistance) predominate (49, 295, 296).

In the European and North American CVD prevention guidelines, the use of coronary artery calcium (CAC) with a computed tomography (CT) scan is endorsed for risk clarification in patients at intermediate risk (5–7% CVD event in 10 years) who may be good candidates for preventive therapies (295, 297). A normal coronary calcium score is zero (0), indicating no detectable calcific plaque and suggesting a very low risk for cardiac events. A score of 1–10 is minimal disease, indicating a low risk. Score of 11–400 is mild to moderate atherosclerosis and confers moderate risk whereas those >400 show extensive plaque and are at high risk of future CVD events. It has been shown that the CAC score helps to identify intermediate to high-risk women, particularly around menopause, who may be overlooked (294, 298, 299). It was previously reported that 30% of women with a history of preeclampsia had higher CAC scores at the age of 50 compared with controls (300). In addition, a recent study showed that women with a history of preeclampsia have larger noncalcified fibrous fatty plaque volumes compared with controls. Yet, in women with POI the presence of CAC has not been shown to be elevated (301).

In the Multi-Ethnic Study of Atherosclerosis (MESA) it was found that the prevalence of CAC = 0 was slightly lower among women with early menopause compared with those with menopause >45 years (55.1% vs. 59.7%; $p = 0.04$). In addition, 38.9% of women with menopause <45 years who reported MHT use had a median CAC score of 0.0 (IQR: 0.0–27.0) compared with 50.0% of women without MHT use who had a median CAC score of 0.4 (IQR: 0.0–72.0) (302).

In contrast, in a recent study on women at high familial risk for ovarian cancer, having had a risk-reducing salpingo-oophorectomy before menopause, no long-term adverse effect on the development of CAC was found (303). This supports the reverse causality hypothesis, or bidirectional risk, which postulates that early natural menopause may be the result of cardiovascular risk factors, whereas early surgical (or otherwise iatrogenic) menopause is not (166).

In general, a completely negative (zero) CAC score in women below 60 years of age can be considered to reflect a very low CVD risk. When available, the CAC score can be used in women at intermediate CVD risk to guide the selection of MHT type and dosage, ensuring the choice with the safest profile in terms of CVD risk, particularly in cases of elevated CAC score (49) (Figure 4). When CVD risk factors are present, lifestyle advice and treatment of these risk factors—according to the latest guidelines—should be initiated before or concomitantly with MHT. It is important to note that women with a clustering of CVD risk factors have more vasomotor symptoms (49, 304). Elevated blood pressure for instance may cause a variety of symptoms, such as palpitations, hot flushes, headaches, chest pain, pain between the shoulder blades, tiredness and sleeping disturbances, that can be sometimes mistakenly attributed to menopause alone (305, 306). Adequate treatment of blood pressure and other risk factors, if present, reduces these symptoms and facilitates the use of lower doses of MHT. Tailoring MHT treatment on the basis of individual CVD risk assessment requires collaborative care between primary care physicians, cardiologists, internists and gynecologists.

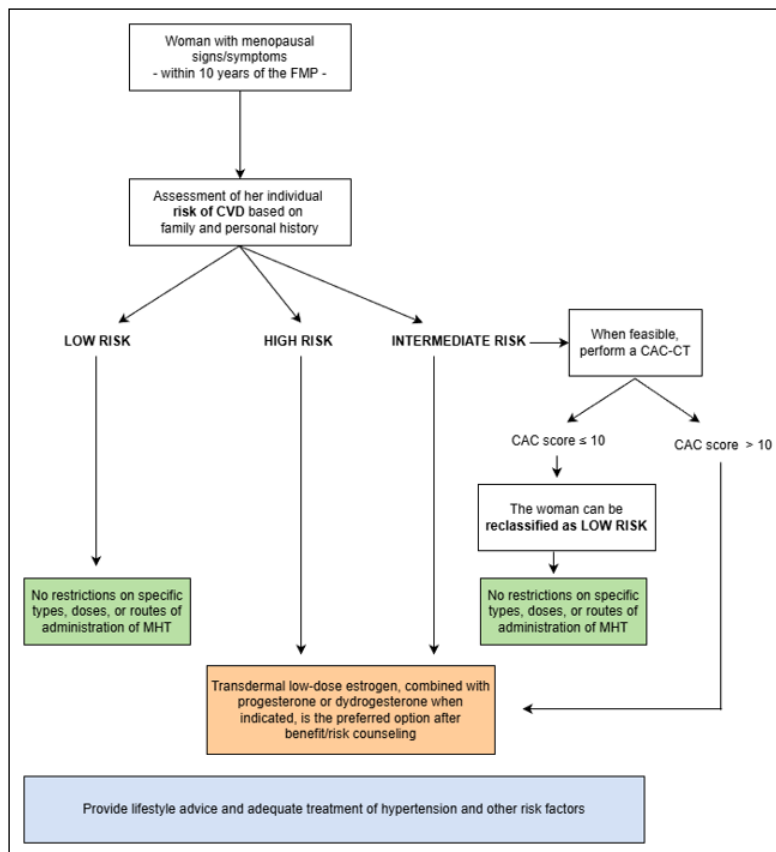


Figure 4 Algorithm for Menopausal Symptom Management and Cardiovascular Risk-Based Decision-Making.

FMP = final menstrual period;
CVD = cardiovascular disease;
CAC = coronary artery calcium;
CT = computed tomography;
MHT = menopause hormone therapy.

MHT IN PATIENTS WITH ESTABLISHED ISCHEMIC HEART DISEASE

The majority of women between 40 and 70 years of age with stable coronary artery disease have ischemia with non-obstructive coronary disease (INOCA) and may demonstrate vascular dysfunction in the epicardial and/or microvascular coronary arteries (307, 308). However, the original WHI report showing an increase in IHD events included predominantly obstructive coronary disease and elderly women (>65 yrs). In more recent randomized as well as registry and meta-analysis data, MHT appears to be more beneficial than harmful when initiated below 60 years of age or within 10 years of onset of menopause (9, 140, 221). In particular, transdermal estrogens may have a beneficial effect in these women, including an increase in insulin sensitivity, improvement of the lipid profile and body composition, decrease in blood pressure in case of drospirenone-containing regimens, and a direct vasodilatory and anti-inflammatory effect (49). In addition, improvements in population diagnoses and treatment control of hypertension, hyperlipidemia and diabetes over the past decades further contribute to a lower risk profile in IHD patients when compared against the time period the WHI studies were done (11).

In middle-aged women, the spectrum of acute coronary syndrome more frequently includes MIs with non-obstructive coronary arteries (MINOCA) and spontaneous coronary artery dissection (SCAD) than the classical obstructive MIs. Both MINOCA and SCAD are associated with fewer traditional CVD risk factors than obstructive CAD, which is more common in men and in women over 65 years of age. A 2001 retrospective study on women above 55 years admitted to hospital with MI showed that current MHT users, at the time of the event, had a significantly lower mortality rate compared with non-users (309). On the other hand, as there is evidence that SCAD may be triggered by sex-hormones, particularly in pregnancy, there is not enough safety data for MHT use in this group. Data on MHT safety after MI are still limited. In one randomized, placebo-controlled trial of secondary prevention in over 1,000 post-menopausal women (age 50–69) low dose unopposed oral estrogen had no significant increase in reinfarction, or cardiac or all-cause mortality (310).

Based upon the one study above, the British Menopause Society recently advised that cessation of MHT may not be necessary in women after MI and they recommend to provide expert care in the choice of hormone type, dose and route of administration, together with the co-administration of a statin and other secondary prevention measures (311) (Figure 5).

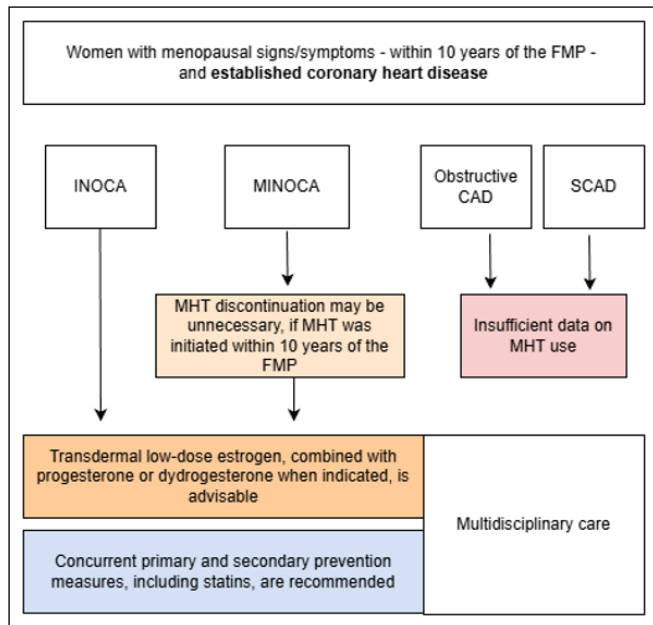


Figure 5 Algorithm for Menopausal Symptom Management in patients with established Ischemic Heart Disease.

FMP = final menstrual period; CAD = coronary artery disease; INOCA = ischemia with nonobstructive CAD; MINOCA = myocardial infarction with nonobstructive CAD; SCAD = spontaneous coronary artery dissections.

CONCLUSIONS

This joint FIGO-WHF position paper underscores that menopause should be considered a ‘window of opportunity’ for comprehensive individual risk profile evaluation and a personalized, evidence-based use of MHT, especially in women with CVD risk factors or established coronary heart disease.

We recommend embedding routine cardiovascular screening within midlife wellness checks and menopausal care pathways, while ensuring equitable access to evidence-based interventions and strengthening interdisciplinary education across relevant specialties

Future efforts should focus on generating high-quality, prospective data on the safety and long-term effects of MHT in women with different cardiovascular risk profiles. Research should also aim to develop sex-specific cardiovascular risk assessment models that integrate reproductive history and both endogenous hormonal milieu and exogenous hormone exposures. In addition, the development of more accessible and less costly alternatives to CAC-CT would facilitate assessment of coronary health status in routine clinical care worldwide. The key messages for clinicians are summarized in the [Box](#).

- ✓ **Menopause** is a critical **window of opportunity** for cardiovascular risk assessment and preventive care in women
- ✓ **Early initiation** of MHT—within 10 years from the final menstrual period or in women under 60—definitively alleviates menopausal symptoms and **may be associated** with a reduction in all-cause mortality and a reduction in the composite outcome of death from cardiovascular causes
- ✓ **Risk stratification** is **essential**: MHT type, dose and route of administration should be **tailored**, based on individual health status, cardiovascular and thromboembolic risk factors. **In women at high risk, transdermal estrogen** combined with **micronized progesterone** or **dydrogesterone**, if the uterus is intact, is the preferred option
- ✓ MHT may also be considered in women with **established ischemic heart disease (IHD)** for menopausal symptom relief, following careful counseling and shared decision making. Transdermal low-dose estrogen combined, if indicated, with micronized progesterone or dydrogesterone is the suggested option
- ✓ **Concurrent** use of standard **primary** and **secondary CVD prevention measures**, including statins and blood pressure management, is fundamental when prescribing MHT to women with CVD risk or established IHD
- ✓ **Multidisciplinary collaboration** between specialists (cardiologists, gynecologists, internists, primary care providers) enhances care for complex cases and supports safer MHT use in women at high CVD risk

Box Key Messages for Clinicians

FIGO and WHF advocate for international collaboration to address evidence gaps and improve cardiovascular risk assessment, thereby further harmonizing standards of post-reproductive healthcare globally. Through their joint leadership, they remain committed to taking actions that help translate scientific knowledge into everyday clinical practice and ultimately improve women's health-span.

COMPETING INTERESTS

Chiara Benedetto has acted in an advisory capacity for Gedeon Richter.

Martha Gulati serves on a data monitoring committee for Merck on a lipid lowering agent. MG Serves on advisory boards/consulting for Amgen, Bayer, Boehringer Ingelheim, Regeneron and Pfizer.

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Suvarna S. Khadilkar has no competing interests to declare.

Angela H.E.M. Maas has no competing interests to declare.

Federica Frezet has no competing interests to declare.

Gláucia Maria Moraes de Oliveira has no competing interests to declare.

Andrea R. Genazzani is immediate Past President of the International Society of Gynecological Endocrinology (ISGE) and European Society of Gynecology (ESG), President of the International Academy of Human Reproduction (IAHR), and Executive Director of ISGE. He has lectured and acted in an advisory capacity for Abbott, Astellas, Bayer, Besins, Bionorica, Elea, Gedeon Richter, Secure Pharma, Serelis, and Theramex.

Hema Divakar has no competing interests to declare.

Athena Poppas is a member of WHF board.

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